This pipeline is designed to automate the processing of raw qpcr export files, perform QC and collate into a clean sheet. We have designed it to be used by people with minimal coding experience.

Instructions

1. Install Python and required libraries
   a. Go to https://realpython.com/installing-python/ and install
   b. Open your terminal and run:

```
python3 -m pip install pandas openpyxl
```

   This installs pandas for spreadsheet and table operations and openpyxl for saving Excel (.xlsx) files

2. Create a new directory, [your_project_folder] and add your files using the following structure (don't edit file names):

```
/your_project_folder/
├── qpcr_pipeline.py      <-- (this script)
├── plate_maps.csv        <-- (your exported plate maps, see figure 1)
├── target_mapping.csv    <-- (your target assignments, see figure 2)
└── raw/   <-- (folder with qPCR .CSV files, named according to plate map)
    ├── Run1.csv
    ├── Run2.csv
    └── ...
```

   a. make a new directory within [your_project_folder] called 'raw' and drop in all your raw exported .csv qPCR files. Raw files must be in .csv format.

Raw exported .csv files from each run in a separate folder ‘raw’ inside [your_project_folder]

raw

Name	Date Modified	Size	Kind
ECHH_MYLC1_112024.csv	3/20/25	26 KB	Com....csv)
ECHH_MYLC2_112024.csv	3/20/25	25 KB	Com....csv)
ECHH_MYLC3_121224.csv	3/20/25	28 KB	Com....csv)
ECHH_MYLC4_122024.csv	4/29/25	15 KB	Com....csv)
ECHH_MYLC5_020725.csv	4/29/25	21 KB	Com....csv)

- Run name <u>must be in the top left</u> of each plate map and must <u>match exactly</u> to the corresponding raw .csv file name (see left image).
- Each sample name should be <u>unique</u>, and placed in the respective location on the qPCR plate
- Each plate map should be separated by a single line
- Positive/negative controls should contain POS, NEG or NTC/NEC in name to be recognized as controls

plate_maps.csv contains all the qpcr plate maps for all runs to be analyzed. <u>Do not change file name</u>.

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	ECHH_MYLC1_112024	1	2	3	4	5	6	7	8	9	10	11	12
2	A		1.1_sup	8.7_sup	8.7_sup	12.1_sup	12.1_sup	19.5_sup	19.5_sup	12.5_sup	12.5_sup	NTC	NTC
3	B	1.5_sup	1.5_sup	8.11_sup	8.11_sup	12.6_sup	12.6_sup	19.6_sup	19.6_sup	33.5_sup	33.5_sup	ECHH POS 50,000	ECHH POS 50,000
4	C	1.6_sup	1.6_sup	NEC1	NEC1	16.7_sup	16.7_sup	20.7_sup	20.7_sup	33.6_sup	33.6_sup	ECHH POS 5,000	ECHH POS 5,000
5	D	4.7_sup	4.7_sup	9.7_sup	9.7_sup	16.11_sup	16.11_sup	20.12_sup	20.12_sup	35.7_sup	35.7_sup	ECHH POS 500	ECHH POS 500
6	E	4.12_sup	4.12_sup	9.11_sup	9.11_sup	16.12_sup	16.12_sup	21.1_sup	21.1_sup	35.11_sup	35.11_sup	ECHH POS 50	ECHH POS 50
7	F	6.7_sup	6.7_sup	9.12_sup	9.12_sup	10.6_sup	10.6_sup	21.5_sup	21.5_sup	35.12_sup	35.12_sup	ECHH POS 5	ECHH POS 5
8	G	6.11_sup	6.11_sup	4.11_sup	4.11_sup	33.1_sup	33.1_sup	NEC2	NEC2	36.7_sup	36.7_sup		
9	H	6.12_sup	6.12_sup	20.11_sup	20.11_sup	19.1_sup	19.1_sup	24.1_sup	24.1_sup	36.12_sup	36.12_sup		
10													
11	ECHH_MYLC2_112024	1	2	3	4	5	6	7	8	9	10	11	12
12	A	40.1_sup	40.1_sup	42.12_sup	42.12_sup	49.7_sup	49.7_sup	NEC3	NEC3	122.7_sup	122.7_sup	NTC	NTC
13	B	40.5_sup	40.5_sup	43.1_sup	43.1_sup	49.11_sup	49.11_sup	119.7_sup	119.7_sup	122.11_sup	122.11_sup	ECHH POS 50,000	ECHH POS 50,000
14	C	40.6_sup	40.6_sup	43.5_sup	43.5_sup	49.12_sup	49.12_sup	119.12_sup	119.12_sup	122.12_sup	122.12_sup	ECHH POS 5,000	ECHH POS 5,000
15	D	41.8_sup	41.8_sup	43.6_sup	43.6_sup	109.7_sup	109.7_sup	120.1_sup	120.1_sup	28.7_sup	28.7_sup	ECHH POS 500	ECHH POS 500
16	E	41.11_sup	41.11_sup	NEC4	NEC4	109.11_sup	109.11_sup	120.6_sup	120.6_sup	28.11_sup	28.11_sup	ECHH POS 50	ECHH POS 50
17	F	41.12_sup	41.12_sup	48.7_sup	48.7_sup	109.12_sup	109.12_sup	121.7_sup	121.7_sup	28.12_sup	28.12_sup	ECHH POS 5	ECHH POS 5
18	G	42.7_sup	42.7_sup	48.11_sup	48.11_sup	114.7_sup	114.7_sup	121.12_sup	121.12_sup	125.7_sup	125.7_sup		
19	H	42.11_sup	42.11_sup	48.12_sup	48.12_sup	114.12_sup	114.12_sup	NEC5	NEC5	125.11_sup	125.11_sup		
20													
21	ECHH_MYLC3_121224	1	2	3	4	5	6	7	8	9	10	11	12
22	A	1.7_WS	1.7_WS	16.1_WS	16.1_WS	NEC3	NEC3	48.1_WS	48.1_WS	62.12_WS	62.12_WS	NTC	NTC
23	B	4.7_WS	4.7_WS	19.1_WS	19.1_WS	35.1_WS	35.1_WS	49.7_WS	49.7_WS	63.7_WS	63.7_WS	ECHH POS 50,000	ECHH POS 50,000
24	C	5.1_WS	5.1_WS	20.7_WS	20.7_WS	36.7_WS	36.7_WS	53.8_WS	53.8_WS	NEC4	NEC4	ECHH POS 5,000	ECHH POS 5,000
25	D	6.7_WS	6.7_WS	21.7_WS	21.7_WS	40.1_WS	40.1_WS	54.7_WS	54.7_WS	68.7_WS	68.7_WS	ECHH POS 500	ECHH POS 500

Figure 1: Instructions for preparing plate_maps.csv file.

target_mapping.csv maps target names to each fluorophore. <u>Do not change file name</u>.

- Assign your qPCR targets their relevant fluorophore.
- Ensure these fluorophores names are consistent with the exported qPCR .csv files.
- The Run pre-fix can be used if you are running different assays with the same fluorophore. If not needed, leave as DEFAULT
- Each fluorophore can only be assigned one target per plate (RunPrefix). Use unique sample names to indicate target if needed.

	A	B	C
1	RunPrefix	Fluorophore	Target
2	RP-SC2	FAM	SC2
3	RP-SC2	CY5	RNAseP
4	ECHH	FAM	EBV
5	ECHH	CY5	HSV
6	ECHH	HEX	CMV
7	ECHH	ROX	HHV6
8	DEFAULT	FAM	Unknown
9	DEFAULT	CY5	Unknown

Figure 2: Instructions for preparing the target_mapping.csv file.

3. Run the script: double click on the shortcut

Output files

Files saved into [your_project_name] directory:

- your_project_name _qpcr_combined_results.csv
    - all qpcr results collated together according to plate and target maps)
- your_project_name _qpcr_clean_summary.xlsx
    - All cleaned qPCR data that passed QC. Results are collapsed by Sample name to give one row per Sample, per target. If enabled, correction to an internal control is completed at this stage.
    - All inconclusive results flagged and excluded
- your_project_name _qpcr_exclusions.xlsx
    - All samples which were marked as ‘Inconclusive’